Data Analytics Insights Report: Metro Interstate Traffic

Prepared for the Smart-City Mobility Team

Executive summary

Hourly traffic on this interstate averages about 3,290 vehicles, but it swings from 0 to 7,280 vehicles per hour, and the time of day explains most of that swing. Weather and temperature matter far less. Congestion (more than 5,500 vehicles per hour) occurs in about 15% of hours. In 2017, the 4 pm hour was the only hour of the day whose average traffic exceeded that threshold, so congestion is concentrated and can be managed with targeted measures rather than all-day ones.

Measure	Result
Hours analysed	40,575 (one row per hour, from 48,204 raw rows)
Average / median traffic	3,291 / 3,427 vehicles per hour
Standard deviation	1,985 (about 60% of the mean)
Range	0 to 7,280 vehicles per hour
Congested hours (> 5,500)	6,107 (15.05%)
Traffic categories	Low 64.2%, Medium 20.7%, High 15.1%

Data and approach

The analysis uses the Metro Interstate Traffic Volume dataset (hourly traffic with weather and holiday information). Data was prepared in SQLite and Power Query: dates were parsed as day/month/year, repeated hours (7,629 extra rows with different weather descriptions) were removed, ten impossible 0 Kelvin temperatures were set to blank, and temperature was converted to Celsius. SQL was used for yearly trends, holidays, descriptive statistics and probability. Descriptive statistics and correlation were also calculated in Power BI using DAX measures, and the Power BI dashboard presents the findings interactively.

Key findings

1. Time of day drives traffic more than anything else. In 2017, average traffic is lowest at about 390 vehicles at 3 am, rises sharply from 5 am to 4,820 by 7 am, stays near 4,300 to 4,800 through midday, and peaks at 5,834 at 4 pm, about 15 times the overnight low. Hours cluster at two levels (below about 1,000 and around 4,500 to 5,000), so the overall average describes few actual hours.

2. Underlying demand was stable, then grew in 2017. Raw yearly totals are misleading because the data has gaps (for example, 2015 starts in June and 2014 ends in August). Average hourly volume stayed between about 3,200 and 3,300 until 2016, then rose to 3,377 in 2017, a like-for-like increase of about 5.6% on 2016. Traffic is lowest in winter and highest from spring to autumn.

3. Holidays and disruptions are clearly visible. Traffic on New Year's Day and Labor Day is roughly 28% to 35% below the surrounding week. 25 December is the quietest day in each year (averages of 1,574, 1,453 and 1,890 vehicles per hour in 2015, 2016 and 2017). On 23 and 24 July 2016, traffic dropped to 277 and 928 vehicles per hour against about 3,300 normally, which looks like a closure or sensor outage.

4. Weather has a modest effect. Cloudy hours have the highest average traffic (3,617) and fog the lowest among conditions with enough data (2,724), a gap of 893 vehicles per hour. Congestion occurs in 13.2% of clear hours against 17.1% of cloudy hours (odds ratio 0.74). Weather and congestion are not independent, but the differences are a few percentage points, and part of the gap reflects time of day, because cloudy hours include more daytime traffic.

5. Temperature has only a weak positive link. The correlation between temperature and traffic is r = 0.14, so temperature explains about 2% of the variation. Hours at 25°C and above average about 4,400 to 4,800 vehicles, against about 3,000 to 3,300 for most colder hours, and the coldest hours (below −20°C) are the quietest. Traffic still ranges from near zero to about 7,000 at almost every temperature.

Implications for the mobility team

- **Target the peaks.** Focus signal timing, ramp metering and driver messaging on the morning rise (6 to 9 am) and the afternoon peak (3 to 6 pm), especially around 4 pm.
- **Schedule works overnight.** In 2017, average traffic was under 1,000 vehicles per hour from midnight to 5 am, the best window for maintenance. Holidays are another low-demand opportunity, with traffic about 30% lighter.
- **Build forecasts around time and calendar.** Use hour of day, season and holidays as the primary inputs, and treat weather and temperature as secondary adjustments (cloudy or hazy conditions and warm days of 25°C and above show somewhat higher flows).
- **Plan for warm-weather demand.** Warm days carry noticeably higher average flows, so consider this when planning events and summer operations, while remembering the link is weak.
- **Improve data quality and monitoring.** Add automatic alerts for zero or near-zero traffic (as in July 2016), remove duplicate records, validate sensor readings (0 Kelvin temperatures and one implausible 9,831 mm rain value), and close the coverage gaps. Only 2013, 2016 and 2017 are close to complete years.

Limitations and next steps

The results show associations, not causes: time of day and season may influence both weather and traffic. Missing periods limit year-to-year comparisons, and some hours carry several weather labels, of which only one was kept. Recommended next steps are to model traffic with hour and day-type controls, add data on incidents, roadworks and events, and keep the Power BI dashboard updated so the team can filter by hour, weather and traffic category.